

NAJMEH NAYYER

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RESEARCH INTERESTS

Combining experimental and theoretical approaches to elucidate neural principles underlying behavior and cognition, particularly perception and decision-making, with the eventual goal of applying this knowledge to AI.

EDUCATION

B.Sc., Psychology, University of Isfahan, Isfahan, Iran Sep 2024 - Present
Current GPA: 18.41/20 (3.89/4)
Relevant Courses: *General Statistics, IT in Psychology, Research Methods, and Cognitive Psychology.*

SKILLS

Programming: Python and MATLAB
Packages: NumPy, Pandas, SciPy, Matplotlib, Seaborn, PsychoPy, MNE, Scikit-learn, EEGLAB
Other Software: MS Office (PowerPoint, Word, Excel), SPSS
Research Skills: Statistical Analysis, Experimental Design, Data Visualization, Signal Processing (Training)
Soft Skills: Project Management, Event Management, Mentoring, Leadership, Self-Study
Languages: Persian (Native), English (Advanced)

EXPERIENCE

RESEARCH EXPERIENCE

Independent Researcher Sep 2025 - Present
Supervisor: Prof. Maryam Vaziri-Pashkam, University of Delaware

- Behavioral study on object recognition using object kinematograms.
- Formed the study hypothesis, organized the literature review and analyzed the initial pilot data.

Project Assistant Apr 2025 - Aug 2025
Supervisor: Prof. Behrad Noudoost, University of Utah [GitHub Link](#)

- Intraocular transfer of color adaptation under varying ocular conditions.

MENTORING EXPERIENCE

Mentoring Intern, Cognitive Neuroscience Competition for High School Jan 2026 - Present
Iranian Neuroscience Society, Tehran, Tehran, Iran
Study Mentor for the National University Entrance Exam Mar 2025
Saramaden e Rah e Danesh Institute (Notruphil), Hamedan, Iran

- Mentored high school students preparing for the exam and provided academic guidance and support.

PRESENTATIONS

Vahabi, F., Honarmand, F., **Nayyer, N.** (2025). *Chromatic Threshold Stability In The Yellow–Blue Axis*. Poster presented at the Basic and Clinical Neuroscience Congress, Tehran, Tehran, Iran ([Link](#))

SELECTED PROJECTS

Analysis of Natural Object Dataset EEG | Python Ongoing

- Transformed data with Morlet wavelets and implemented supervised machine learning algorithms to compare activity in different classes of stimuli.

Python Bootcamp Projects ([GitHub Link](#)) Ongoing

- Developed 10+ interactive Python projects including games (Snake, Pong, Turtle Crossing) and utilities (Coffee Machine inventory system, Flashcard app), demonstrating OOP, GUI design, and data management.
- Built functional applications spanning game logic, inventory simulation, and study tools using Python fundamentals including loops, conditionals, classes, and event-driven programming.

Analysis of Psychology Class of 2028 Social Status Dataset | Python ([GitHub Link](#)) Ongoing

- Used NumPy, Pandas, SciPy, Matplotlib, and Plotly library to preprocess and analyze a survey from our sociology class.

Gender Inequality Attitudes Questionnaire | SPSS Jul 2026

- Co-designed and validated a 30-item Likert-scale questionnaire for a Psychometrics course project.

- Analyzed data from 50+ participants, performing reliability analysis (Cronbach's $\alpha = 0.957$, Spearman-Brown = 0.978), indicating excellent internal consistency.
 - Calculated descriptive statistics, standardized Z/T scores, and multiple cut-off score ($M + 2SD = 88.44$).
- Memory-Guided Saccade Task | MATLAB (Project Link)

Jun 2026
- Analyzed multi-unit spikes (25 neurons) and LFP from macaque FEF during a memory-guided saccade task. (8 angles, 2 radii) Plotted raster and PTSH.
 - Decoded target location from population firing rates using linear SVM and characterized LFP power spectra using multitaper method.
 - Computed mutual information and AUC to quantify neural encoding of angle and radius across task phases.
- Statistical Analysis Projects | SPSS

2025
- Two SPSS-based statistical analysis projects for the Practical Statistics and IT in Psychology course.
 - Used descriptive statistics, correlation, linear regression, t-tests, ANCOVA, MANOVA, etc.

AWARDS & EXTRACURRICULAR

- Brainies Book Club Founder and Leader (Link)

Jan 2026 - Present
- Presented “A Thousand Brains: A New Theory of Intelligence” by Jeff Hawkins to +10 students.
- Dean’s List, University of Isfahan, Isfahan, Iran

1st Sem- Present
- Vice President, Psychology students' association of University of Isfahan, Isfahan, Iran

Jun 2025 - Present
- Top Rank in the National Entrance Exam, Ministry of Science, Research and Technology, Tehran, Iran

2024
- Ranked 1135 (top 0.03%) among 348,917 participants nationwide which resulted in a full-ride scholarship.

SELECTED CERTIFICATES & OPTIONAL COURSES

- Supervised Machine Learning: Regression and Classification, Andrew Ng, DeepLearning.AI (Coursera)

Nov 2025
- Principles of Neuroscience, Behrad Noudoost, Isfahan University of Medical Sciences

May 2025
- Data Structures, Charles Severance, University of Michigan (Coursera)

May 2025
- AI for Everyone, Andrew Ng, DeepLearning.AI (Coursera)

Jan 2025
- Understanding the Brain: Neurobiology of Everyday Life, Peggy Mason, University of Chicago (Coursera)

Dec 2024
- Introduction to Data Analysis Using Excel, Sharad Borle, Rice University (Coursera)

Nov 2024